PIZZA SALES DATA ANALYSIS

1. Project Overview

The Pizza Sales Analysis project aims to analyze transactional sales data from a pizza store to uncover meaningful business insights. The objective is to evaluate sales performance, identify customer purchasing patterns, and determine key drivers of revenue.

Using SQL for data analysis and Power BI for visualization, the project provides actionable insights that support better decision-making in areas such as marketing strategies, inventory planning, and operational efficiency.

This analysis is aligned with the business goals defined in the Business Requirements Document (BRD), which focuses on identifying key performance indicators, sales trends, and product performance.

2. Dataset Description

The dataset contains transactional records of pizza sales.

Dataset Details

- Total Records: 48,620
- Total Columns: 12

Key Data Fields

The dataset includes the following important attributes:

- order_id
- pizza_id
- pizza_name
- quantity
- pizza_size
- pizza_category
- total_price
- order_date
- order_time

This dataset enables analysis of sales performance, customer preferences, and time-based trends.

3. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Total Revenue - Total revenue generated from pizza sales.**

Results | Messages

	Total_Revenue
1	817860.05083847

2. **Average Order Value - Average amount spent per order.**

Results | Messages

	Avg_order_Value
1	38.3072623343546

3. **Total Pizza Sold - Total number of pizzas sold across all orders.**

Results | Messages

	Total_pizza_sold
1	49574

4. **Total Order - Total number of customer orders placed.**

Results | Messages

	Total_orders
1	21350

5. **Average Pizza per Order - Average number of pizzas ordered per order.**

Results | Messages

	Avg_Pizzas_Per_order
1	2.32

6. **Daily Trend for Total Orders – The number of orders was analyzed for each day of the week.**

Results | Messages

	order_day	Total_orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

7. **Monthly Trend for Total Orders - Monthly sales trends were evaluated to understand seasonal patterns.**

Results | Messages

	Month_Name	Total_Orders
1	July	1935
2	May	1853
3	January	1845
4	August	1841
5	March	1840
6	April	1799
7	November	1792
8	June	1773
9	February	1685
10	December	1680
11	September	1661
12	October	1646

8. **Percentage of Sales by Pizza Category – Calculated share of pizza category sales in terms of percentage**

Results | Messages

	pizza_category	PCT
1	Classic	26.9059602306976
2	Chicken	23.9551375322885
3	Veggie	23.6825910258677
4	Supreme	25.4563112111462

9. **Percentage of Sales by Pizza Size - Calculated share of pizza size sales in terms of percentage**

Results | Messages

	pizza_size	Total_Sales	PCT
1	L	375318.701004028	45.8903330244889
2	M	249382.25	30.492044420599
3	S	178076.49981308	21.7734684107037
4	XL	14076	1.72107684995364
5	XXL	1006.6000213623	0.123077294254725

10. **Top 5 Best Sellers by Revenue, Quantity, and Total Orders**

Results | Messages

	pizza_name	Total_Revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

Results | Messages

	pizza_name	Total_Quantity
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

Results | Messages

	pizza_name	Total_Orders
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Pizza	2273
5	The Thai Chicken Pizza	2225

11. Bottom 5 Best Sellers by Revenue, Quantity, and Total Orders

Results | Messages

	pizza_name	Total_Revenue
1	The Brie Carre Pizza	11588.4998130798
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Pizza	15277.75
4	The Mediterranean Pizza	15360.5
5	The Spinach Pesto Pizza	15596

Results | Messages

	pizza_name	Total_Quantity
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

Results | Messages

	pizza_name	Total_Orders
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Spinach Supreme Pizza	918
4	The Calabrese Pizza	918
5	The Chicken Pesto Pizza	938

4. Data Preparation and Dashboard Development in Power BI

To create meaningful visualizations and improve time-based analysis, additional calculated columns and measures were created within Power BI using DAX (Data Analysis Expressions). These enhancements allowed for better aggregation, filtering, and visualization of sales trends.

4.1 Calculated Columns (DAX)

Several calculated columns were added to the dataset in the Power BI Editor to support time-based analysis and improve the usability of dashboard visuals.

Column Name	Purpose
Day Name	Extracted the weekday name from the order date column (Monday, Tuesday, etc.)
Order Day	Extracted first 3 letters from Day Name column (MON, TUE, etc.)
Day Number	Represents the numeric order of the weekday to ensure proper sorting in visuals (SUN = 1, MON = 2)
Month Name	Extracted the month name from order date column (January, February, etc.)
Order Month	Extracted first 3 letters from Month Name column (JAN, FEB, etc.)
Month Number	Numeric representation of the month to maintain chronological order in charts

These calculated columns help ensure that charts and visuals are sorted correctly and enable better analysis of daily and monthly sales patterns.

4.2 DAX Measures Created

To support KPI visualization in the dashboard, the following DAX measures were created:

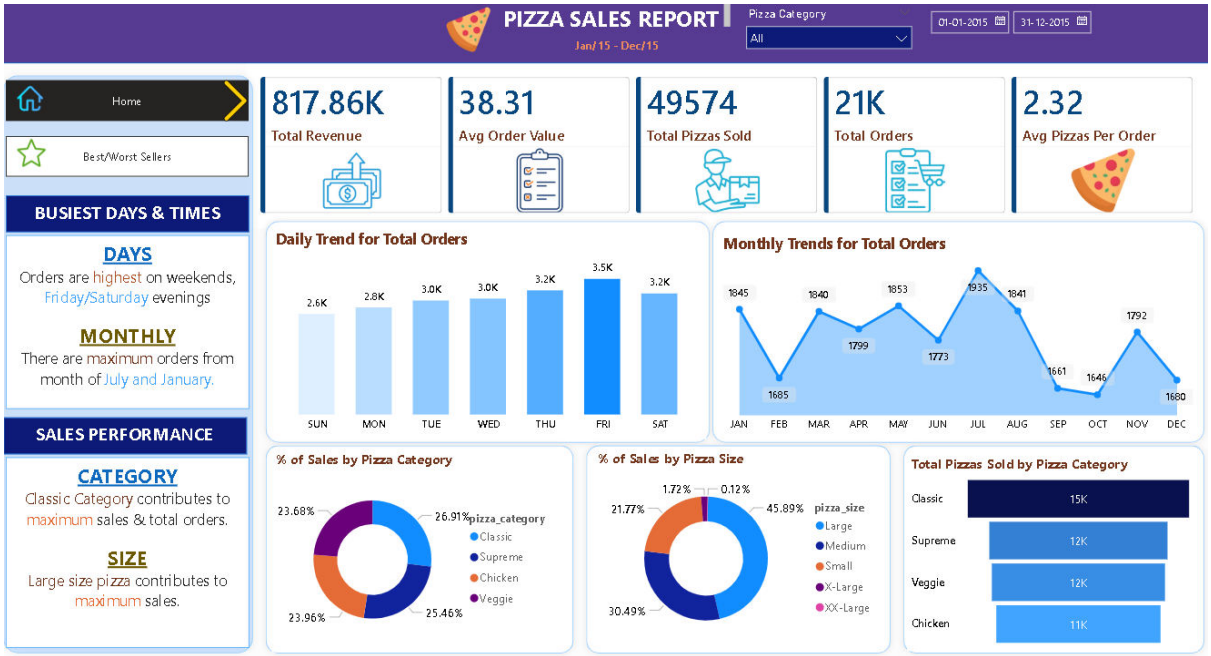
Measure	Description
Total Revenue	Total revenue generated from pizza sales
Total Pizzas Sold	Total number of pizzas sold
Total Orders	Total number of orders placed
Average Order Value	Average amount spent per order
Average Pizzas per Order	Average number of pizzas ordered per transaction

These measures dynamically calculate key business metrics and allow interactive filtering within the dashboard.

4.3 Power BI Dashboard

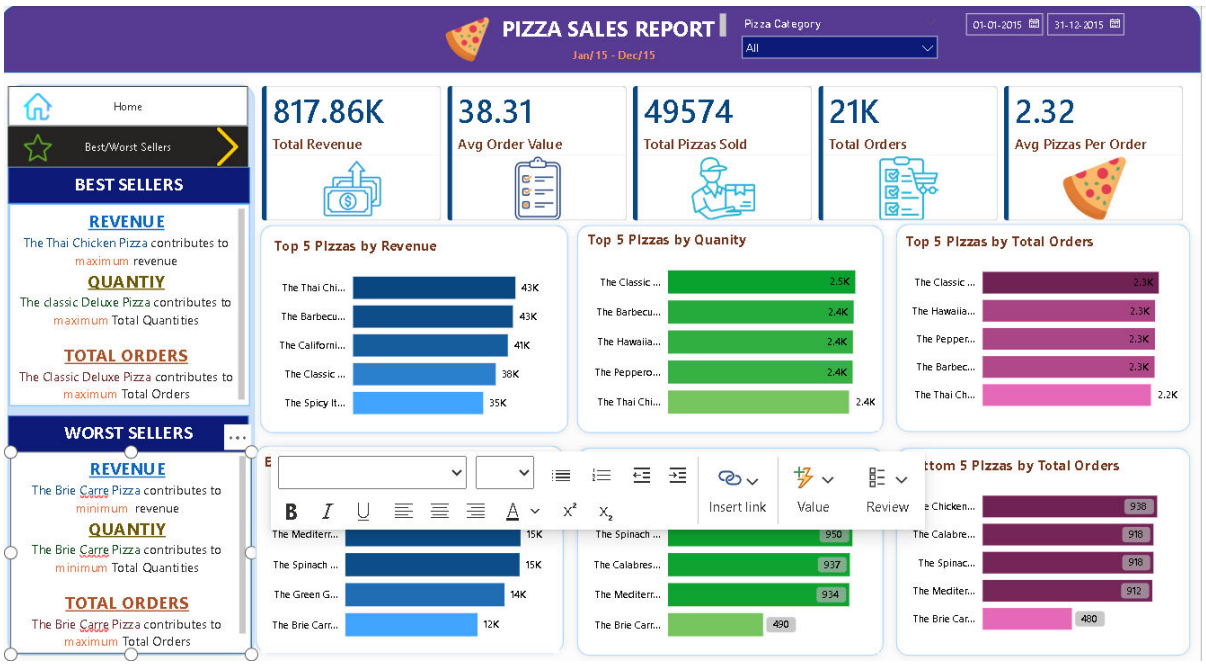
We built an interactive dashboard in POWER BI to present insights visually.

Created a **Home dashboard** containing daily and monthly trends of pizza sales and the percentage of pizza sales by category size.



This dashboard provides a high-level overview of business performance and allows users to quickly identify trends and patterns.

Created **Best and Worst sellers dashboard** depicting the top 5 and bottom 5 pizzas by revenue, quantity, and total orders



These visuals help management identify high-performing products to promote and low-performing items that may require improvement or removal from the menu.

5. Key Business Insights

From the analysis, several important insights were identified:

- The Classic pizza category generates the highest sales.
- Large-sized pizzas contribute the largest share of revenue.
- Thai Chicken Pizza generates the highest revenue.
- Classic Deluxe Pizza leads in terms of quantity sold and total orders.
- Brie Carre Pizza performs the worst across revenue, quantity, and orders.
- Most sales occur on Thursdays, Fridays, and Saturdays.
- Peak sales months include July, May, January, August, and March.

6. Business Recommendations

- **Focus Marketing on High-Performing Products**
  Promote popular pizzas such as Thai Chicken Pizza and Classic Deluxe Pizza through marketing campaigns and featured menu placements.

- **Optimize Menu Strategy**
  Evaluate low-performing items such as Brie Carre Pizza and consider improving, repositioning, or removing them from the menu.

- **Improve Inventory Planning**
  Stock more ingredients required for high-demand pizza categories and sizes to avoid shortages during peak hours.

- **Plan Staffing Based on Demand**
  Increase staffing during high-demand days (Thursday–Saturday) and peak hours to ensure faster service.

- **Use Data-Driven Promotions**
  Plan promotional offers during slower days or months to balance demand and increase overall revenue.